

BENXI, China

WMO: 543460

Lat: 41.3067N

Lon: 123.7753E

Elev: 185

StdP: 99.12

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.1	-19.9	-27.6	0.3	-17.4	-25.5	0.4	-16.1	6.2	-5.7	5.3	-5.5	2.1	90	0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.7	31.9	22.2	30.5	21.9	29.3	21.6	25.0	29.0	24.1	28.0	23.4	27.1	2.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.7	19.0	27.5	22.9	18.1	26.4	22.2	17.3	26.0	77.5	29.1	73.9	28.2	71.2	27.1	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.3	5.3	4.6	-25.0	34.1	2.9	2.0	-27.1	35.6	-28.8	36.7	-30.5	37.8	-32.6	39.3		
			-25.3	25.8	2.9	1.0	-27.3	26.5	-29.0	27.1	-30.6	27.6	-32.7	28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	-10.6	-6.0	1.7	10.3	16.8	21.7	24.6	23.6	18.1	10.3	0.7	-7.8	
	DBStd	12.89	4.94	5.27	4.85	4.49	3.47	2.68	2.09	2.58	3.23	4.53	5.73	5.26	
	HDD10.0	2286	638	449	260	51	1	0	0	0	1	53	282	551	
	HDD18.3	4046	897	682	516	242	71	4	0	1	43	252	528	809	
	CDD10.0	1806	0	0	2	60	212	350	454	420	243	61	4	0	
	CDD18.3	525	0	0	0	1	23	104	196	163	36	1	0	0	
	CDH23.3	4204	0	0	0	14	247	834	1606	1266	232	6	0	0	
	CDH26.7	1161	0	0	0	1	46	254	477	356	27	1	0	0	
Wind	WSAvg	2.2	2.0	2.2	2.5	2.7	2.5	2.2	2.0	1.8	2.0	2.1	2.1	2.0	
Precipitation	PrecAvg	806	7	12	21	47	65	115	203	184	58	50	30	14	
	PrecMax	1252	27	40	78	105	176	203	463	455	136	136	78	35	
	PrecMin	540	0	1	0	1	19	38	91	77	5	6	6	1	
	PrecStd	160	6	11	16	26	36	43	70	87	28	28	18	8	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.3	9.2	16.7	25.4	30.1	33.7	34.0	33.3	29.0	24.0	17.0	7.9	
		MCWB	1.1	3.7	7.5	13.2	17.0	19.3	23.6	23.9	20.4	15.5	11.2	5.1	
	2%	DB	1.7	6.3	13.7	22.3	27.9	31.3	32.1	31.0	27.2	21.5	14.6	5.1	
		MCWB	-1.2	1.7	6.4	11.7	15.9	19.2	23.4	23.3	18.8	14.2	9.5	2.1	
	5%	DB	-0.3	4.6	11.5	20.2	26.2	29.5	30.5	29.8	25.9	19.7	12.2	2.8	
		MCWB	-3.0	0.8	5.0	10.8	15.3	19.0	23.0	22.7	18.0	13.2	7.9	0.0	
	10%	DB	-2.4	2.8	9.5	18.3	24.3	27.7	29.3	28.7	24.5	17.8	10.1	0.8	
		MCWB	-4.8	-0.7	3.8	9.7	14.3	18.8	22.7	22.3	17.5	12.3	6.2	-1.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.7	4.9	9.3	15.5	19.3	22.5	26.6	25.9	22.4	17.1	12.2	5.5
			MCDB	3.7	8.0	14.8	21.8	26.5	27.8	31.1	30.3	26.3	22.1	15.9	7.9
2%		WB	-1.0	2.4	7.3	13.3	17.7	21.4	25.3	24.9	21.0	15.5	9.9	2.3	
		MCDB	1.3	5.5	11.9	19.8	24.9	26.7	29.7	28.6	24.9	19.7	14.0	4.8	
5%		WB	-2.9	1.0	5.7	11.9	16.6	20.7	24.4	24.2	19.8	14.3	8.2	0.3	
		MCDB	-0.5	4.0	10.7	18.9	23.5	26.1	28.5	27.9	23.6	18.6	11.8	2.4	
10%		WB	-4.7	-0.5	4.2	10.6	15.6	20.0	23.7	23.5	18.7	13.1	6.5	-1.5	
		MCDB	-2.5	2.7	9.0	16.7	22.2	25.5	27.5	27.1	22.9	16.9	9.7	0.8	
Mean Daily Temperature Range		5% DB	MDBR	10.8	10.4	9.3	10.5	11.0	9.6	7.7	8.4	10.7	10.8	9.4	9.6
			MCDBR	12.0	11.1	12.2	13.8	14.5	13.6	10.4	10.0	12.3	13.0	11.7	10.4
	MCWBR		9.5	7.6	6.9	6.6	5.8	4.7	3.6	3.6	4.8	6.4	7.7	8.3	
	5% WB	MCDBR	11.3	9.5	10.5	10.9	11.4	9.3	7.7	7.4	9.4	10.7	10.3	9.8	
		MCWBR	9.2	7.0	6.7	5.8	5.3	3.8	3.3	3.4	4.5	5.9	7.7	8.3	
Clear-Sky Solar Irradiance	taub	0.378	0.445	0.505	0.602	0.642	0.716	0.692	0.601	0.530	0.543	0.461	0.396		
	taud	2.000	1.845	1.786	1.649	1.612	1.538	1.626	1.798	1.892	1.777	1.896	1.987		
	Ebn at Noon	721	722	730	691	677	629	639	686	703	617	615	656		
	Edh at Noon	121	165	197	242	257	277	252	206	175	176	135	114		
All-Sky Solar Radiation	RadAvg	2.37	3.20	4.27	5.00	5.67	5.10	4.58	4.60	4.36	3.25	2.25	1.93		
	RadStd	0.15	0.22	0.28	0.34	0.45	0.54	0.39	0.41	0.30	0.20	0.24	0.15		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page